Utrecht University Repository

Title	Work Time Reduction: A Critical Analysis of the Main Arguments for Adopting a 4-Day Workweek
Authors	Oliveira, Ana; Cruz, Sofia Alexandra; Couto, Ana Isabel; Cerdeira, Jorge; Parente, Cristina; Gonçalves, Carlos Manuel
Publication Date	2025-06-12
Link	https://dspace.library.uu.nl/handle/1874/482965
Citation	Oliveira, A, Cruz, S A, Couto, A I, Cerdeira, J, Parente, C & Gonçalves, C M 2025, Work Time Reduction : A Critical Analysis of the Main Arguments for Adopting a 4-Day Workweek. in C Machado (ed.), Human Resource Management in a Disrupted World. 1 edn, Lecture Notes on Multidisciplinary Industrial Engineering, vol. F528, Springer, pp. 333-355. https://doi.org/10.1007/978-3-031-89948-5_14
Versions / License	Publisher version
Rights	https://www.uu.nl/en/university-library/license-and-reuse-conditions

Chapter 13
Work Time Reduction: A Critical Analysis of the Main Arguments for Adopting a 4-Day Workweek

Ana Oliveira, Sofia Alexandra Cruz, Ana Isabel Couto, Jorge Cerdeira, Cristina Parente, and Carlos Manuel Gonçalves

Abstract Over the past five years, the private sector, policymakers, and civil society have developed a growing interest in reduced working hours. The rapid growth of 4-day workweek pilots in multiple firms across different countries has led to considerable scholarly attention regarding this topic. However, most of this discussion is taking place outside of the academic domain. The lack of representativeness of academic literature on the ongoing debate implies that a comprehensive scientific review of the arguments about the 4-day workweek is still missing. This chapter fills this gap by critically assessing the most frequently cited arguments for adopting shorter weeks—increasing productivity, improving well-being, correcting gender imbalances, and reducing environmental impact—showing that some of these arguments lack sound scientific support. It frames the 4-day workweek discussion in the general debate about reducing working time in different periods in time and discusses the results of the main 4-day workweek pilots carried out over the most recent years.

Keywords Working time · Work time reduction · 4-day workweek · 4-day workweek arguments

A. Oliveira
Utrecht School of Economics, Utrecht University, Utrecht, Netherlands

S. A. Cruz · A. I. Couto (✉)
Faculty of Economics of University of Porto, Institute of Sociology of the University of Porto, Porto, Portugal
e-mail: aicouto@fep.up.pt

A. I. Couto
CEOS.PP, ISCAP, Polytechnic of Porto, Porto, Portugal

J. Cerdeira · C. Parente · C. M. Gonçalves
Faculty of Arts and Humanities of University of Porto, Institute of Sociology of the University of Porto, Porto, Portugal

J. Cerdeira
Center for Economics and Finance at the University of Porto, Porto, Portugal

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

C. Machado (ed.), *Human Resource Management in a Disrupted World*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-3-031-89948-5_14

13.1 Brief Introduction

Shorter working weeks and, in particular, implementing a 4-day workweek (4DWW) of 30–32 working hours as a norm have received much attention in the wake of COVID-19, the emergence of the so-called "Great Resignation" phenomena, and growing difficulties in hiring and retaining workers.[1] Academics, political parties, policymakers, and think tanks have been increasingly interested in such debate, reflected in extensive media coverage, new research publications, and a growing number of management-led trials.

In this chapter,[2] we aim to understand how the claims for the (4DWW) have emerged in the context of the evolution of working time since the early twentieth century and how they are framed within the wider debate on working time reduction. We critically review the main arguments for the 4DWW, confronting them with empirical evidence and compiling the results of the most well-known 4DWW pilots.

This chapter has three aims. First, it frames the 4DWW discussion within the general debate on reducing working time in different periods. Second, it explores the most frequently cited arguments in favour of the 4DWW—increasing productivity, improving well-being, correcting gender imbalances, and reducing environmental impact (Campbell, 2023; Eurofound, 2024; Harper & Martin, 2018; Jahal et al., 2023; Veal, 2023)—and confronts them with the existing robust empirical evidence on the effects of reducing working time. Finally, it summarises the design and reported outcomes of the foremost 4DWW pilots conducted over the most recent years and the limitations of these experiments.

13.2 Reducing Working Time in Different Periods in Time

Despite the recent buzz around the 4DWW, this is not the first time that it has been debated. The first reported shortened week dates back to 1930, when 8-h shifts at Kellogg's manufacturing plant in Michigan were reduced to 6 working hours a day, with no wage cuts (Hunnicutt, 1992). Kellogg's primary motivation for introducing shorter shifts was to create more jobs in Battle Creek during the Great Depression. Moreover, the company tried to retain its workforce in a highly routinised job by increasing workers' motivation and satisfaction.[3]

Kellogg's management-led experience of shorter regular working hours became a relevant case study in a context in which the 40-h working week had recently been implemented in only some large plants and a few US states. The path to a statutory maximum of 40 "normal" weekly hours was still under construction. The Fair Labor

[1] https://time.com/charter/6241211/four-day-workweek/.

[2] The research that supports this chapter was conducted in July 2023.

[3] Changes in management in the late 30 s initiated four decades of constant attempts to increase the number of daily working hours back to eight. In 1984, under threat of reallocation and job displacement, workers who were still working under six-hour shifts resigned (Hunnicutt, 1992).

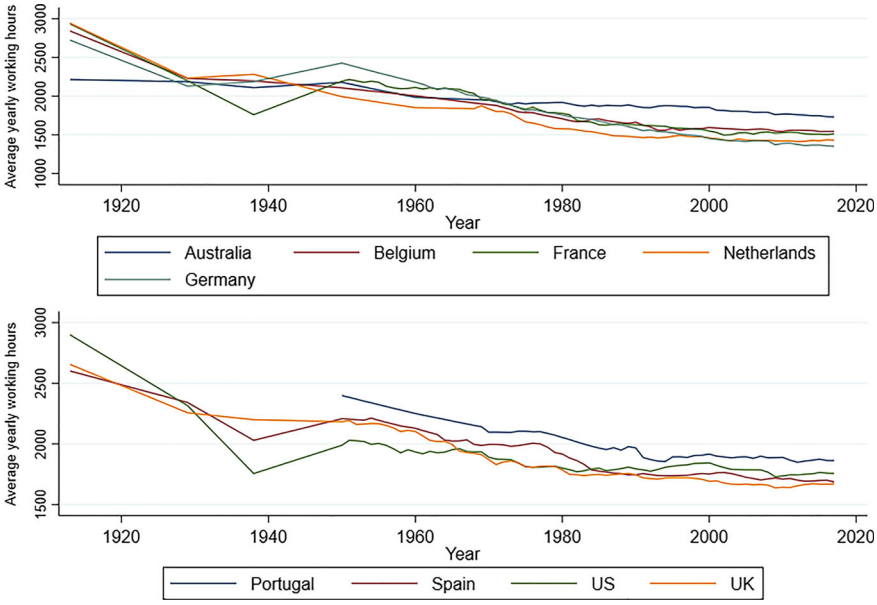

Fig. 13.1 Average yearly working hours. *Note* Based on Giattino et al. (2020) and our world data

Standards Act of 1940 made the 40-h week the law in the USA. By then, the average working week was still 46 h long in the UK (Bank of England, 2023) and over 47 h in manufacturing plants in the Netherlands and Sweden (Gilmore, 2021). Workers in these countries had to wait until the mid-fifties to reach 40 weekly working hours. Figure 13.1 shows the evolution of yearly working hours for a selection of European countries and the US.

Working hours fell significantly before the Second World War: in countries such as the Netherlands and France, average yearly hours fell by 750 between 1913 and 1929. Equally significant changes occurred between 1929 and 1938. The ILO Hours of Work (Industry) Convention, 1919 (No.1), ratified in the late 1920s and early 1930s by countries such as Belgium (1926), France (1927), Portugal (1928), Spain (1929) or Canada (1935), represented an important milestone towards the eight-hour working day in the industrial sector during this period.

The war and post-war efforts restored longer working periods. Of all the countries considered, only the Netherlands and Belgium registered a decrease in the number of hours worked between 1938 and 1950. In all other countries, those increased, sometimes reversing all the progress made in the previous decade.

The pre-war downward trend in working time was resumed in the fifties. Between 1950 and 1970, several European countries registered double-digit reductions in yearly working hours: 10% in Belgium and Spain, 11% in France, 14% in the UK, and 19% in Germany (Fig. 13.2). The increase in the number of vacation days, sick leave, and parental leave were the main drivers of this decrease over this period,

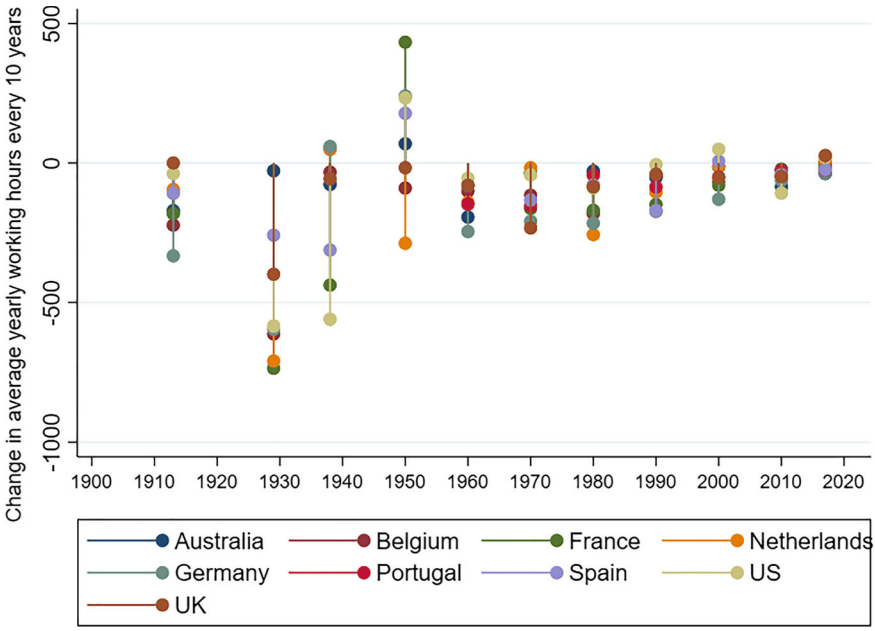

Fig. 13.2 10-year change in average yearly working hours. *Note* Based on Giattino et al. (2020) and our world data

although shorter working days have also been reported, especially in the US (Costa, 2000).

In the 1970s, some US companies tried compressed 4-day working weeks (40 weekly hours in 4 days). A meta-analysis of the literature examining the impacts of compressed working weeks revealed that workers were generally satisfied with the new working schedule, despite reporting increased fatigue (Moores, 1990). Similar pilots were replicated in the UK and Australia. However, compressed working weeks did not become the norm in all companies and there is no evidence that pilot companies adopted this schedule after the experiment ended (Veal, 2023).

As in the 1920s and 1930s, the potential impact on employment (work-sharing) was the most referred argument in favour of reduced working hours during the 1970s, when the US unemployment rate reached 8.8% (US Bureau of Labor Statistics, 2023). Work sharing was again a popular argument to introduce the 35-h standard working week in France in the mid-nineties, motivating a series of economic papers to determine the relationship between employment and working time reductions.[4]

From the 1990s onwards, the decrease in yearly working hours stalled. In the UK, where yearly working hours fell by 17% between 1960 and 1990, only a 3% reduction was registered in the following thirty years; in the US, the 2.5% reduction between

[4] The results are ambiguous, but reducing working hours may increase employment under certain conditions. A compilation of empirical evidence is described in Estevão et al. (2008).

1990 and 2017 contrasted with the 7%-change in the previous period; in Germany, where the most recent change is considerable (−14.5%), it was still far below the change in working time achieved between the 1960s and the 1990s (−27%) (Giattino et al., 2020).

The literature documents two episodes of national reductions in the statutory working week. In 1996, the Portuguese Government reduced the standard working week from 44 to 40 h, affecting around 50% of all private sector employees (Raposo & Ours, 2010). The policy aimed converge with the European working time standards and companies had two years to implement the new working schedule. In 1998, the French Government passed a law reducing the 40-h working week to 35 h, which came into force for all companies in 2002. High unemployment and the expectation that the policy would encourage work-sharing were at the core of the political debate in France at the time (Estevão et al., 2008). However, the impact on employment was not statistically significant (Estevão et al., 2008).

Two countries have introduced compressed 4-day workweeks into their labour law. With the Working Time Act of 1997 and amended in 2007, Austria opened up the possibility of extending the daily working time to 10 h a day to accommodate a 4-day workweek. Belgium, in 2001, ruled for flexible schedules in the private sector of a 9-h working day in 4 days (Cabrita & Boehmer, 2016). The most recent campaigns for the 4-day workweek have instead claimed for an effective reduction in the standard working week, in most cases to 30 or 32 h.

13.3 Arguments for a 4-Day Workweek: Discussion and Empirical Evidence

The arguments in favour of the 4-day workweek (4DWW) cover many fields and are ideologically diversified (Boulin, 2024; Burchell et al., 2024; Sanchis, 2023). The more ancient literature placed more emphasis on the revolutionary ideal of human emancipation, considering the 4DWW as a way of freeing people from the burden of work and giving them more time for leisure and creative activities (Hunnicutt, 1992). Proponents of the 4DWW now emphasise other potential outcomes rather than an increase in leisure time (Veal, 2023).

The arguments in favour of implementing the 4-day workweek can be clustered into the following four themes: (i) increasing productivity, (ii) promoting general well-being, (iii) improve gender imbalances and (iv) reducing environmental impact. These arguments will be discussed in detail in the following sub-sections.

13.3.1 Increase of the Productivity

Highly prominent advocates of the 4DWW implementation have defended its role in increasing workers' productivity. In the book "Overworked American: The Unexpected Decline of Leisure", Juliet Schor mentions that "reduced hours are highly correlated with lower emissions, and they also yield improvements in worker well-being, gender equity, and productivity" (Schor et al., 2022); in 2019, the think tank Autonomy states: "there are a number of reasons why a reduction of working hours is desirable and possible with productivity in mind." (Stronge & Harper, 2019); in "Friday is the New Saturday" (2021), Pedro Gomes argues that "the four-day week would reduce this pressure [on mental health] and raise workers' productivity." Similar claims can be found in countless news and reports, often referring to the outstanding results reported after the end of pioneering 4DWW pilots.

In most of these documents, we find references to the impact of long working hours on labour productivity. Long working hours is still common in many countries (ILO, 2023), especially among high-skilled workers (Cortes & Pan, 2019; Kodz et al., 2023). Long working hours are culturally associated with a prototype of a worker who puts their job above everything else, is highly committed to their job, and is highly productive (Kim et al., 2018; Rosengren, 2019; Rutherford, 2001). In certain occupations and workplaces, the ideal worker is always available, can travel and change work locations, and prioritises work over family time and leisure (Weeden et al., 2016; Williams, 2010).

Some authors claim that the association between long working hours and high individual productivity is inaccurate and may involve a performative act to meet hierarchical expectations and seek a career promotion (Heejung, 2022; Weeden et al., 2016). Despite the expectations of employers and employees, empirical evidence shows that increasing individual working hours does not proportionally increase the outcome and decreases hourly productivity (Golden, 2011; Shepard & Clifton, 2000). The negative relationship between long working hours and labour productivity stems from the impact of the former on health outcomes, as well as fatigue and stress (Collewet & Sayermann, 2017; Wong et al., 2019). Rivera et al. (2020) show that people working more than 40 h per week are more likely to suffer from coronary heart disease, and pregnant women are more likely to have premature births and low birth weight. Long working hours are also associated with a higher incidence of musculoskeletal disorders (Johnson & Lipscomb, 2006), cerebrocardiovascular disease (Wong et al., 2019), stress and fatigue (Johnson & Lipscomb, 2006; Pencavel, 2015), depression, anxiety and sleep condition (Afonso et al., 2017; Bannai & Tamakoshi, 2014; Milner et al., 2015; Wong et al., 2019). Long working hours are also related to higher odds of suffering a work-related injury (Dembe et al., 2005; Dong, 2005) and being overweight (Li et al., 2021; Virtanen et al., 2022).

Reducing overtime may have had a positive impact on labour productivity. Firstly, it helps prevent the occurrence of the health hazards described above (Golden et al., 2011; Sparks et al., 2001), thus reducing absenteeism. Secondly, teams whose

members experience less fatigue are, on average, more productive (Shangguan et al., 2021) and less likely to make mistakes at work (Landrigan et al., 2004).

Apart from Schor et al. (2022), Gomes (2021) and Autonomy (2019), many other authors (Barnes, 2002; Chung, 2022; Pang, 2020; Spencer, 2022) reference existent empirical evidence on the effects of reducing long working hours (50 or more hours) to argue for a reduction in the standard working week (35–40 h). Such reference bias is likely related to the scarcity of robust and consensual evidence on the causal relationship between regular working hours and labour productivity. Indeed, many studies that aim to prove causality use industry-level data only, which suffers from endogeneity when measuring both working time and output (Collewet & Sayermann, 2017).

One of the few exceptions is Collewet and Sayermann (2017). The authors look at the productivity variation along an 8-h working day and find that for a 1% increase in working time, individual output increases by only 0.9%. Fatigue explains this loss of productivity. Accordingly, one may expect that reducing the length of the working day increases hourly productivity, but it is unknown whether the same applies to reducing the duration of the working week.

Only four other papers examine the relationship between part-time work and company productivity. The results are contradictory. Specchia and Vandenberghe (2013) and Decicienti et al. (2015) found a negative impact of employing a larger share of part-time workers on the productivity of Belgian and Italian companies, respectively. Garnero et al. (2014) and Künn-Nelen et al. (2013), however, found that Belgian and Dutch part-time workers are relatively more productive than full-timer workers. Across the four studies, results vary by gender, part-time duration, and sector.

In a systematic review of studies examining the impact of reduced working hours on health outcomes (while keeping the same wage and workload), Voglino et al. (2022) show that there is a positive relationship between a shorter working week and working life quality, improved sleep habits and reduced stress. Although these positive outcomes seem to depend on the organisational aspects of the workplace and how leisure time is spent (Voglino et al., 2022), shorter working hours contribute to preventing some of the underlying causes of absenteeism (Van De Poll et al., 2020), thus increasing the number of productive days.

David Spencer (2022) argues against placing productivity at the centre of the 4DWW debate. The author states that whenever working time has been reduced on the initiative of the employers to increase productivity, 4-day workweek experiences have either failed (such as at Microsoft Japan, whose pilot was never extended), or back-fired on workers' well-being by keeping the same workload but reducing the time to complete it and increasing work intensity. Work intensification and higher stress levels were reported in 4DWW trials when resources and workload remained unchanged despite the reduced working time (Delaney & Casey, 2022). Similar outcomes were reported in some British companies participating in the 2022 pilot.[5]

[5] https://www.bbc.com/worklife/article/20230319-four-day-workweek-trial-the-firms-where-it-didnt-work.

The "pure" effect of reducing working hours on company productivity—by reducing unproductive time, reducing fatigue, or introducing more efficient management methods (e.g., shorter meetings)—may be difficult to quantify. The lack of robust evidence indicates this. Comparatively, the effect of shorter statutory working hours on absenteeism and stress-related sick leaves is more debated and the existing literature, while also scarce, is more consensual. Overall, we find that the case for the 4DWW as a means of increasing productivity is not well supported by empirical evidence and is based mainly on the literature focused on long working hours.

13.3.2 Promotion of General Well-Being

In the article "A Social Policy Case for the 4-day workweek" (2021), Chung states that "providing workers more capacity to spend time with others can also help improve the well-being of others—such as their children and other family members". Similar claims can be found on the 4-day workweek Global webpage, while most 4DWW pilots have examined the experiment's impact on well-being indicators. It is, thus, safe to say that concerns about physical and psychological well-being have been at the core of the latest 4DWW debate.

Similarly to the productivity argument, many 4DWW advocates refer to the literature focused on long working hours and their consequences for workers' well-being (Gomes, 2021; Heejung, 2022; Lewis & Stronge, 2021). Based on this literature, they suggest that reducing statutory working hours will prevent the negative well-being outcomes observed when employees work long hours, often assuming a linear relationship between the two.

Zheng et al. (2023) show precisely that the relationship between job satisfaction and working hours is non-linear and point out the existence of heterogeneous effects across genders and types of employment.[6] Dong (2005) and Kamerade et al. (2019) also demonstrate that life satisfaction increases as working hours decrease non-linearly. Although these results may sound trivial, they highlight the importance of treating long and regular working hours differently.

The relationship between long weekly working hours, job satisfaction, and health outcomes is well documented. Some of the extensive literature on long working hours and health outcomes—e.g., cardiovascular disease, sleep quality, obesity, stress, and mental health conditions such as depressive and anxiety symptoms—was already described above. Several other papers also provide evidence of the negative effects of long working hours on overall well-being and jobs (e.g., Anfrave & Charlwood, 2015; Zheng et al., 2023).

Although it is reasonable to expect that fewer working hours increase leisure time and free up time for physical activity, which has been shown to positively

[6] In particular, they show that job satisfaction decreases as the number of hours worked increases for married women (but not for unmarried women) and hat job satisfaction decreases for self-employed women only after 12 daily working hours.

affect people's health, the relationship between shorter statutory working weeks and workers' physical and mental well-being is considerably less documented.[7] A few exceptions report the Swedish experiences of working time reductions in various companies from the private (Schwarz & Hasson, 2011; Schwarz et al., 2008) and the public sectors (Schiller et al., 2017) and well-being outcomes. Health benefits were found whenever physical activity replaced working time (Schwarz et al., 2008), and less stress and longer sleep duration were reported by public servants who benefited from shorter working hours (Schiller et al., 2017).

Lepinteur (2019) evaluates the nationwide reduction of the statutory weekly working hours in France and Portugal in the late nineties. The author shows that implementing a shorter working week had positive, robust, and non-transitory effects on overall job satisfaction due to higher satisfaction with working hours and perceived working conditions. The paper also explores heterogeneous outcomes across sectors and gender: men employed in manufacturing and agriculture were relatively more satisfied with their jobs after France introduced the 35-h working week compared to the remaining workforce; in Portugal, the reduction in weekly working hours seemed to increase job satisfaction in all sectors, particularly among women. Berniell and Bientenbeck (2020) look at the impact of the French reform on health-related outcomes and demonstrate that four years after the reform, workers whose working week was reduced to 35 h were less likely to smoke, had a lower BMI and reported better health than workers who were not subject to the reform.

Despite the focus of the 4DWW literature on the hazards of long working hours for workers' general well-being and satisfaction, the non-linear pattern between working time and wellbeing-related variables confirms the need to use the typical working week as the baseline scenario to test whether a 4DWW improves these variables. Studies such as Lepinteur (2019) and Berniell and Bientenbeck (2020) prove the benefits of reducing working time below standard working hours to these outcomes.

13.3.3 Improvement of Gender Imbalances

In 2019, the World Economic Forum claimed that the 4DWW would be a game changer in achieving gender equality.[8] Women's rights organisations, such as UNICEF, the ILO or the UK Women's Budget Group, have appealed to the 4DWW adoption as a strategy to improve women's access to leisure time and more equal wages. Likewise, prominent 4DWW advocates mention the contribution of shorter working weeks to narrow gender gaps at home and at the workplace. It is safe to

[7] Voglino et al. (2022) conducted a systematic review of research publications on quantitative health outcomes of reducing working hours without loss of income, and the pool of publications was reduced to seven, all documenting experiences in Sweden.

[8] https://www.weforum.org/agenda/2019/01/why-the-four-day-work-week-is-good-news-for-gender-equality.

say that these organisations and personalities strongly believe in changing gendered social norms by changing working schedules.

Gender pay gaps are still significant in most countries, although they have been narrowing in recent decades. In OECD countries, women still earn 11.9% less than men, in EU countries 10%, and in the UK and US 15 and 17%, respectively. Social norms, and large inequalities in the distribution of care and household responsibilities between men and women, not only contribute to the persistence of gender gaps in the labour market but also prevent women's equal participation in several societal dimensions (e.g., men are twice as much involved in politics as women (EIGE, 2021).

Across the EU Member States, women spend an average, of 16.5 h a week on cooking and/or housework outside paid work, 67% longer than men (EIGE, 2021). In 2016, European mothers spent 38.8 h per week on caring and/or educational activities outside paid work, while fathers spent only 20.5 h (EIGE, 2021). In most OECD countries, women combine time spent on unpaid work with 40 weekly hours of paid work. Moreover, in countries where part-time work is prevalent (such as Germany or the Netherlands), women are more likely to be employed part-time.[9] Gender differences in the use of time and the duration of paid work harm women's career prospects by reducing the probability of being promoted, limiting mobility between companies, and benefiting from lower wage gains when switching between companies (OECD, 2021).

Gendered social norms reduce women's full-time employment and affect the relative hourly wages of male and female workers. Some companies disproportionately reward individuals who work long hours and outside regular working hours. First, the gender gaps in hourly wages increase because off-hours are paid at higher rates and men are over-represented among those rewarded (Goldin, 2014). Second, men are more likely to be promoted in these companies (Blau & Kahn, 2013). Third, companies with non-linear hourly wages that promote unpredictable (and deregulated) working hours often employ high-wage professionals such as law firms, financial services, and consulting (Goldin, 2014). As women with children tend to seek flexible and shorter working hours, they are less likely to be employed in these companies, widening the gender pay gap (Card et al., 2016). Overall, time is "a resource to which men have more access than women" (Rutherford, 2001, p. 274), and as organisations demand more of this resource, women's job market prospects are damaged.

Reducing weekly working hours may contribute to improving women's labour market outcomes. First, reducing the gap between part-time and full-time work may reduce the wage penalty associated with part-time work (Budig & England, 2001; Dias et al., 2018). Second, shorter full-time schedules may induce part-time women to move to full-time employment (Spiegelaere & Piasna, 2017) and may reduce the incentives to disproportionately reward men for time availability. Therefore,

[9] In Austria, almost 35% of employed women work less than 30 h a week, while less than 8% of men do; in Belgium, 26% of women and 8.1% of men; in Germany, 34% of women and 8.7% of men; in Netherlands, 51.7% of women work part-time (<30 h), and the rate drops to 8.7% among men (OECD, Incidence of FTPT employment, 2022).

the gender pay gap related to the number of hours worked may be narrowed by implementing a reduced working week. In Portugal, where the working week was reduced from a maximum of 44–40 h between 1996 and 1998, reducing the statutory working hours narrowed the gender employment gap between 0.7 and 1.9% points.[10] Although the gender employment gap in Portugal was already small, due to the low occurrence of part-time employment and the traditionally high female labour market participation rate, reducing the regular working week narrowed the gap even further. Consequently, a more significant impact may be expected in countries where female part-time employment is more prevalent.

Advocates of the 4DWW often refer to its potential to reduce the inequalities in the division of household tasks (4 Day Week Global, 2023; Chung, 2022; Coote et al., 2021). However, survey data descriptions collected after some 4-day-week pilots and more robust qualitative studies show that reducing working time does not necessarily mean a more equal redistribution of care and household tasks between spouses when men spend less time at work. Piasna and Spiegelaere (2021) mention the observation of increased male participation in unpaid work after the French reform from 40 to 35 weekly hours, but to a small extent and without structural changes in social norms. The report on the Icelandic 35-h-week pilot also shows modest results: in some cases, men started participating more in household tasks, but women did not perceive significant normative changes (Haraldsson & Kellam, 2021). No change in the distribution of household labour was found in the recent UK 4-week pilot either: 68% of the survey trial respondents reported no change in this regard (Lewis et al., 2023).

Consistent with these views, Flèche et al. (2020) demonstrate that (1) men who work significantly less than their female partners do not spend much more time on household work than men who work significantly more time than their female partners, and (2) women also spend similar amounts of time on unpaid work whether they work more or less than their partners.

Overall, gender differences in the distribution of time between paid work, unpaid work, and leisure time within the household seem to depend much more on social norms and individual views of these social norms (Flèche et al., 2020). Therefore, despite the high hopes that shortening the working week will change social norms and ease women's burden of unpaid work, there is little empirical evidence to support this. While social norms are remarkably persistent (Giuliano, 2017), there are public policies able to change them, with significant intergenerational effects.[11]

[10] Controlling for age, tenure, education, and 1-digit occupations. The first estimate (0.7 p.p.) is obtained by adding firm fixed effects and the second by adding industry fixed effects. Data from *Quadros de Pessoal,* authors' analysis.

[11] Farré et al. (2023) analyse the introduction of paternity leave in Spain and found that children born after the reform were more likely to have gender egalitarian attitudes. Bünning (2015) found that German fathers who took at least two months' leave were more involved in subsequent childcare.

13.3.4 Reduction of Environmental Impact

Perspectives on the 4-day week and its environmental impact are also divergent. Orthodox macroeconomics wrongly assumes that economies will grow indefinitely, because ecosystems, although self-regenerating, also deteriorate when overexploited (Pullinger, 2014; Stagl, 2014). Thus, it is understood that the environmental impact can be reduced if fewer working hours result in a decrease in demand for transport and consumption of water and light in the workplace (Ashford & Kallis, 2013). Furthermore, if productivity gains are invested in leisure instead of capital, there can be a fall in output growth, with a consequent reduction in consumption, while at the same time increasing well-being, in a perspective known as sustainable degrowth (Fitzgerald et al., 2018; Knight et al., 2012; Pullinger, 2014). In this way, it is increasingly assumed that technological change alone will not achieve sustainable consumption, so the path to sustainability must be based on stabilising consumption by reducing working hours (Marston, 2016; Schor et al., 2022). This idea is supported by two effects: the scale effect, which refers to the impact of working hours on the size or scale of the economy through GDP growth, and the composition effect, which measures the influence that the availability of time has on the carbon intensity of household consumption (Fitzgerald et al., 2018). On the one hand, the scale effect means that, without mechanisms to reduce working hours, longer working hours generate a higher level of economic production (GDP) and, consequently, income and consumption (Fitzgerald et al., 2018; Knight et al., 2012). This creates a cycle called 'work and spend' which shapes workers' behaviour to lead them to consume more, because productivity increases translate into production and consumption, with the aggravating factor that consumption is environmentally degrading (Knight et al., 2012). On the other hand, the composition effect implies a decision-making component at the family level that incorporates income and time availability (Fitzgerald et al., 2018). In this sense, it states that families with shorter working hours and more leisure time will be more inclined to adopt less ecologically intensive lifestyles, because productivity increases are channelled into more free time rather than more income and consumption, unlike families with a shortage of time (Knight et al., 2012; Pullinger, 2014; Schor et al., 2022).

The objections raised regarding the relationship between the 4-day workweek and its environmental effects refer to the uncertainty of the environmental impacts. If, on the one hand, capital and energy were to replace workers, the environmental damage would increase. On the other hand, it is not known whether the extra leisure time will be redirected to less resource-intensive activities and, therefore, it is not certain that there will be a reduction in consumption (Kallis et al., 2013; Pullinger, 2014; Schor et al., 2022; Worstall, 2019). How leisure is used depends on people's consumption habits, their interests and priorities, the friends they have, the opportunities that are available and the money they have at their disposal (Coote et al., 2021; Schor et al., 2022). Given that there is still an association between success and greater economic ownership and, at the same time, between greater economic ownership and more leisure, the tendency is to buy more things, which increases

the consumption of resource-intensive products (Coote et al., 2021; Worstall, 2019). Therefore, environmental benefits should not be taken for granted unless measures are taken to control resource-intensive consumption (Kallis et al., 2013; Pullinger, 2014; Stagl, 2014).

13.4 Empirical Evidence on 4-Day Workweek Pilots

Although the first experiments to implement a shorter working week belong back to the 1930s—with the six-hour shifts in Kellogg's manufacturing plant—the number of private companies piloting a 4DWW has boomed in the most recent years. Examples of completed pilots can be found for instance in Australia, Belgium, Brazil, Canada, Japan, New Zealand, Portugal, South Africa, Spain, UK, and the US. Some of these trials began as a single-company initiative—e.g., Perpetual Guardian in New Zealand or Microsoft Japan, while others were simultaneous experiments in multiple companies that voluntarily joined collective trials, such as in the UK and Portugal. In all cases, private companies self-selected for the experiment, adjusting the basic setting—reducing the number of hours and the number of weekly working days with no reduction in wages—to a working schedule fitting its needs the best (e.g., all workers in the company have the same day-off or half of workers has Mondays off, and the remaining half has Fridays).

Some of these pilots have been subject to significant media exposure, expanded the number of advocacy groups and 4DWW supporters, and have influenced other experiments. It is the case of the pilot run at Perpetual Guardian, a 250-worker company in New Zealand, which tested a shorter working week in 2018 and adopted it after the end of the trial, and whose CEO leads the international advocacy group "4 Day Week Global". It is also the case of the pilot in Iceland, held in the public sector, which did not implement 4DWW but only a reduction of working time from 40 to 35–36 weekly hours (distributed across five days); remarkably, the Icelandic experience is widely used as an example of the 4-day workweek success both by the media, advocacy organisations (e.g., 4-day workweek Global, Autonomy) and some policymakers. In 2019, the women's organisation Femma Wereldvrouwen conducted a 12-month trial, reducing the working time to 30 h a week (Mullens et al., 2021). More recently, 61 UK-based companies participated in the UK pilot over six months. The results of this pilot, which attracted a lot of media coverage, are described by the UK-based think tank Autonomy.[12] Moreover, in 2022, the organisation 4-day workweek Global conducted pilots in 33 companies operating entirely remotely or based in the US, Ireland, UK, Australia, and New Zealand.

Most of the reported 4DWW trials reveal outstanding results on productivity, workers' well-being, and job satisfaction. Perpetual Guardian reported a 20% productivity increase and a significant improvement in workers' perception of work-life

[12] The results are in: the UK's four-day week pilot". Autonomy, 2023. Available at https://autonomy.work/portfolio/uk4DWWpilotresults/.

balance (Cuello, 2023). Microsoft Japan, frequently referred to as an example of a successful 4DWW pilot, reported a 40% productivity increase (Chappell, 2019). Revenues also rose significantly in several companies participating in the 4-day workweek Global multi-country trial: compared to the period before the trial, 20 participating companies had, on average, registered a 37% increase in revenues (Schor et al., 2022). The UK report uses the same metric and states that 34 participating organisations (out of 61) had met, on average, a 35%-expansion in revenues (Lewis et al., 2023).

Most workers covered by the pilots reported lower stress levels, lower incidence of burnout symptoms, better sleep, and less fatigue (Booth, 2019; Lewis et al., 2023; Schor et al., 2022). Among other impacts, workers stated that it was easier to balance work and family life (Schor et al., 2022), and they were spending more time with their children (Delaney & Casey, 2022; Lewis et al., 2023). Consistent with other findings described in the previous section, surveyed men did not spend more time performing household chores when they started having a third day off (Lewis et al., 2023).

These and other results are described in further detail in Table 13.1, which holds a comprehensive overview of the main 4DWW pilots reported in publicly available sources in English, until July 2023. This excludes the experiment developed in Microsoft Japan, for which no results are consistently reported.

The outstanding outcomes described in the abovementioned reports contrast with concerns raised about the countervailing impacts of these experiments. Delaney and Casey (2022) conducted a qualitative case study on the Perpetual Guardian pilot and assessed the implications of this experiment as a management-led initiative, mainly motivated to increase labour productivity and workforce engagement (Delaney & Casey, 2022). The authors found that, despite the initial workers' involvement to reflect upon organisational changes (e.g., how to measure productivity, how to innovate working processes), the weight of maintaining the same workload to hold a shorter week—which required increasing individual productivity and work pace—counterbalanced the overall positive attitudes towards the trial. In particular, some staff felt compelled to perform some tasks over the day off (or being available to do it if necessary), increasing stress levels. Moreover, Delaney and Casey (2022) observed stricter management monitoring of employees and performance measurement and the absence of a "social rights and collective interests' perspective" (p. 185).

Spencer (2022) expresses similar heed. In particular, the author is concerned that if work time reduction is motivated by productivity gains, the 4DWW can backfire on workers and hinder wage progression, job quality among the low-skilled, and companies' rent-sharing. These adverse effects may be amplified if a 4DWW is achieved through compressed weeks or wage cuts.[13]

The results of the 4DWW pilots described above have also been questioned based on their robustness and external validity. Cuello (2023) screens three of the most cited pilots by the OECD quality assessment metric, which sets the standards for policy

[13] As Telefonica, a Spanish communication company, attempted to test an opt-in 4DWW with a 12% wage cut and found little support among employees.

Table 13.1 4DWW reported pilots

Pilot	Country	No. of companies	Year and pilot duration	Performance metric	Health and wellbeing	Job and life satisfaction	Work-life balance and use of time	Other reported impacts	References
Perpetual guardian	New Zealand	1	2018, 2 months	Productivity: +20%	Reduction of work stress: from 45 to 38% Some individuals experienced greater workloads		Increase in work-life balance scores: from 54 to 78%		Booth (2019), Delaney (2018)
4-day workweek Global	Ireland, US, UK, Australia, New Zealand, remote	33	2022, 6 months	Revenue change from comparison to the same period in 2021: +37.6% (weighted average among companies) Revenue, change from trial start and endpoint comparison to trial period: +37.6% (weighted average)	Reduction of work stress: −0.21 on a 1–5 scale Reduction in burnout symptoms: −0.44 on a 1–5 scale; Increase in time spent doing physical exercise: more 23 min/week Reduction in fatigue and sleep problems: −0.38 and −0.33 on a 1–4 scale	Increase in job satisfaction by 0.28 on a 1–10 scale Increase in life satisfaction: +0.89 on a 0–10 scale	Improvement in combining paid work with care duties: +0.72 on a 1–5 scale Improvement in combining paid work with social life: +0.78 on a 1–5 scale No change in time spent on caring responsibilities and housework No change in the gendered division of non-paid labour	Reduction in time spent communing: 1 h per week Increase in international traveling, energy use, and leisure travel	Schor et al. (2022)

(continued)

Table 13.1 (continued)

Pilot	Country	No. of companies	Year and pilot duration	Performance metric	Health and wellbeing	Job and life satisfaction	Work-life balance and use of time	Other reported impacts	References
UK Pilot	UK	61	2022, 6 months	Revenue change compared to the same period in 2021: +34.5% (weighted average across companies) Revenue, change from trial start and endpoint comparison to trial period: + 1.4% (weighted average)	Reduction of work stress: −0.33 on a 1–5 scale Reduction in burnout symptoms: −0.46 on a 1–5 scale Reduction in fatigue and sleep problems: 40% of workers report fewer sleep problems Negative emotions are experienced less frequently	Increase in job satisfaction by 0.57 on a 0–10 scale	Improvement in combining paid work with care duties: +0.82 on a 1–5 scale Improvement in combining paid work with social life: +0.88 on a 1–5 scale No change in the gendered division of non-paid labour	Increase in international traveling, energy use, and leisure travel Reduction in childcare costs	Lewis et al. (2023)
Femma Wereldvrouwen	Belgium	60	2019, 1 year		Reduction in stress and pressure in employees' private lives Higher levels of life satisfaction		Women spent more time on household-related activities but experienced less stress while performing those Some women became less satisfied with the division of unpaid labour	Some employees experienced less open interaction moments with colleagues Workers became more aware of their working time usage	Mullens and Ignace (2023) Mullens et al. (2021)

evaluation (OECD, 2023). Cuello categorizes the limitations with the methodologies adopted in each experiment into four groups: (1) concerns with addressing causality (omitted variable bias and selection bias), (2) issues with collecting accurate data (e.g., low-quality administrative data), (3) limited transparency (e.g., incomplete statistical information, misuse of accessible summaries), and (4) incomplete information on sample and intervention characteristics (e.g., lack of sample representativeness, small sample size). The author finds that all pilots analysed suffer severe methodological flaws across all four categories. In particular, the pilots failed to collect quality data, report all statistical information collected, disclose pre-registered plans, run robustness checks, gather a representative sample, address heterogeneous effects, analyse data from non-compliers, and provide information on costs and scalability of the intervention (Cuello, 2023). Some of these limitations imply, for instance, that macroeconomic contexts—which may be correlated with some of the experiment outcomes (e.g., productivity)—are not considered at all, and there is an overselling of positive results, while (partially) ignoring more modest (or inexistent) results.

Overall, one may conclude that future 4DWW pilots should rely on robust experimental design, collect high-quality administrative data, ensure the robustness of the results, and provide adequate reporting.

13.5 Discussion and Conclusion

Working hours significantly decreased in the decades following the Second World War. Extending paid holidays, sick leave, and parental leave, as well as an effective reduction in weekly working hours, contributed to lower yearly working hours. The downward trend stalled in the 1990s, and only a few countries reduced the standard weekly working hours. Sectoral collective bargaining agreements and company-specific practices have sometimes led to lower full-time schedules. At the same time, in some high-wage countries, workers—especially women—have opted for part-time work, thereby reducing their payslips proportionally.

The debate on reducing working weeks intensified after the COVID-19 pandemic. Changes in how (and from where) people work, the introduction of technological applications that make teamwork more effective (even at a distance) and can monitor employees more closely, and increasing difficulties in retaining workers led some companies to test 4-day working weeks. In most cases, there is an effective reduction in working hours (rather than compressed weeks) and no wage cuts. The acknowledgment that men and women used their time in lockdown differently highlighted the persistence of gender norms that have led to gender inequalities in the distribution of unpaid work and labour market outcomes. Labour organisations, women's rights advocacy groups, left-wing think tanks and politicians have, thus, campaigned for the 4DWW.

The arguments most often used in favour of the 4DWW have been grouped under four themes: increasing productivity, improving well-being, correcting gender imbalances, and reducing environmental impact. We noted that although the demand for a

4DWW is for shorter statutory working hours, its advocates often refer to the consequences of working long hours to make the case for the 4DWW. Existing literature proves that the hazards of working overtime may not necessarily apply to standard working hours. Therefore, we argue that the impact of shorter working hours should be assessed regarding the existing statutory schedules. When researching empirical evidence on the positive effects of shorter standard working weeks on productivity, workers' well-being and gender inequality, we conclude that it is somewhat limited and, sometimes, poorly robust.

The claims on the impact of the 4DWW on productivity gains are susceptible to the choice of working hours—standard or including overtime—as the baseline. The importance of this argument in the media debate requires a better understanding of the effective change in companies' outcomes when employees work fewer hours. Moreover, some studies indicate that making the 4DWW implementation dependent on productivity gains may constrain the well-being benefits of shorter working hours. Further studies assessing these countervailing effects of performing the same tasks in shorter periods are lacking.

We reviewed the reports describing the 4DWW pilots. Only a few trials were assessed and their experimental design suffered from significant methodological limitations. Often, the results of these trials are reported in the media without sufficient evaluation of their actual impact and external validity.

Overall, we conclude that the discussion on working time reductions is opportune, considering the stagnation of hours worked, the recent productivity gains driven by technological change and the positive effects on workers' well-being. However, 4DWW trials should be based on sound and robust policy evaluation methods and identify the advantages, disadvantages and challenges of implementing a 4DWW in the workplace.

Funding This work was funded by the Foundation for Science and Technology under Project UID/00727/2020. Oliveira thanks to Instituut Gak for support. Couto also thanks to the Portuguese national funds through FCT – Fundação para a Ciência e Tecnologia, under the project UIDB/05422/2020.

References

4 Day Week Global. (2023). From https://www.4dayweek.com

Afonso, P., Fonseca, M., & Pires, J. F. (2017). Impact of working hours on sleep and mental health. *Occupational Medicine, 67*, 377–382.

Anfrave, D., & Charlwood, A. (2015). What is the relationship between long working hours, over-employment, under-employment and the subjective well-being of workers? Longitudinal evidence from the UK. *Human Relations, 68*(9), 1491–1515.

Autonomy. (2019). The shorter working week: A radical and pragmatic proposal. https://autonomy.work/wp-content/uploads/2019/03/Shorter-working-week-docV6.pdf

Ashford, N., & Kallis, G. (2013). A four-day workweek: A policy for improving employment and environmental conditions in Europe. *The European Financial Review*, 53–58. www.europeanfinancialreview.com

Bank of England. (2023). Average weekly hours worked in the United Kingdom [AWHWUKA]. Retrieved from FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/AWHWUKA

Bannai, A., & Tamakoshi, M. (2014). The association between long working hours and health: A systematic review of epidemiological evidence. *Scandinavian Journal of Work Environment and Health, 40*(1), 5–18.

Barnes, A. (2002). *The 4 day week*. Piatkus.

Berniell, I., & Bientenbeck, J. (2020). The effect of working hours on health. *Economics and Human Biology, 29*, 100901.

Blau, F. D., & Kahn, L. M. (2013). Female labor supply: Why is the US falling behind? *American Economic Review, 103*(3), 251–56.

Booth, R. (2019). *Four-day week: Trial finds lower stress and increased productivity*. From The Guardian. https://www.theguardian.com/money/2019/feb/19/four-day-week-trial-study-finds-lower-stress-but-no-cut-in-output

Boulin, J.-Y. (2024). Semaine de Quatre Jours: Une Préfiguration de l'avenir de Temps de Travail? *Futuribles, 4*(461), 57–76.

Budig, M. J., & England, P. (2001). The wage penalty for motherhood. *American Sociological Review, 66*, 204–225.

Bünning, M. (2015). What happens after the 'Daddy Months'? Fathers' involvement in paid work, childcare, and housework after taking parental leave in Germany. *European Sociological Review, 31*(6), 738–748.

Burchell, B., Deakin, S., Rubery, J., & Spencer, D.A. (2024). The future of work and working time: Introduction to special issue. *Cambridge Journal of Economics, 48*(1), 1–24. https://doi.org/10.1093/cje/bead057

Cabrita, J., & Boehmer, S. (2016). *Working time developments*. Eurofound.

Campbell, T. T. (2023). The four-day work week: A chronological, systematic review of the academic literature. *Management Review Quarterly, 74*(1), 1791–1807.

Card, D., Cardoso, A. R., & Kline, P. (2016). Bargaining, sorting, and the gender wage gap: Quantifying the impact of firms on the relative pay of women. *The Quarterly Journal of Economics, 131*(2), 633–658.

Chappell, B. (2019). 4-day workweek boosted workers' productivity By 40%, Microsoft Japan Says. *NPR*.

Chung, H. (2022). A social policy case for a four-day week. *Journal of Social Policy, 51*(3), 551–566.

Collewet, M., & Sayermann, J. (2017). Working hours and productivity. *Labour Economics, 47*, 96–106.

Coote, A., Harper, A., & Stirling, A. (2021). *The case for a four-day week*. Wiley.

Cortes, P., & Pan, J. (2019). When time binds: Substitutes for household production, returns to working long hours, and the skilled gender wage gap. *Journal of Labour Economics, 37*(2), 351–398.

Costa, D. L. (2000). The wage and the length of the work day> from the 1890 to 1991. *Journal of Labor Economics, 18*(1), 156–181.

Cuello, H. (2023). *Assessing the validity of four-day week pilots*. JRC working papers series on social classes in the digital age.

Decicienti, F., Grinza, E., & Vannoni, D. (2015). The impact of part-time work on firm total factor productivity: Evidence from Italy. *IZA DP No. 9463*.

Delaney, H. (2018). *Perpetual guardian's 4-day workweek trial: Qualitative research analysis*. The University of Auckland.

Delaney, H., & Casey, C. (2022). The promise of a four-day week? A critical appraisal of a management-led initiative. *Employee Relations: The International Journal, 44*(1), 176–190.

Dembe, A., Erickson, J., Delbos, R., & Banks, S. (2005). The impact of overtime and long work hours on occupational injuries and illnesses: New evidence from the United States. *Occupational and Environmental Medicine, 62*(9), 588–97.

Dias, M. C., Joyce, R., & Parodi, F. (2018). *Wage progression and the gender wage gap: The causal impact of hours of work*. The Institute for Fiscal Studies.

Dong, X. (2005) Long workhours, work scheduling and work-related injuries among construction workers in the United States. *Scand J Work Environ Health, 31(5*, 329–35.

EIGE. (2021). *Gender statistics database*. From https://eige.europa.eu/gender-statistics/dgs/about

Estevão, M., Sá, F., & Petrongolo, B. (2008). The 35-hour workweek in France: Straightjacket or welfare improvement? *Economic Policy, 23*(55), 417–463.

Eurofound. (2024). Working time reduction with a focus on the four-day week: Literature review. *European foundation for the improvement of living and working conditions.*

Farré, L., Felfe, C., González, L., & Scneider, P. (2023) Changing gender norms across. *IZA DP No. 16341*.

Fitzgerald, J. B., Schor, J. B., & Jorgenson, A. K. (2018). Working hours and carbon dioxide emissions in the United States, 2007–2013. *Social Forces, 96*(4), 1851–1874.

Flèche, S., Lepinteur, A., & Powdthavee, N. (2020). Gender norms, fairness and relative working hours within households. *Labour Economics, 65*, 101866.

Garnero, A., Kampelmann, S., & Rycx, F. (2014). Part-time work, wages, and productivity: Evidence from Belgian matched panel data. *Industrial and Labor Relations Review, 67*(3), 926–954.

Giattino, C., Ortiz-Ospina, E., & Roser, M. (2020) *Working hours.* From OurWorldInData.org. https://ourworldindata.org/working-hours

Gilmore, O. (2021). The working week in manufacturing since 1820. In *How was life? Volume II: New perspectives on well-being and global inequality since 1820.* OECD Publishing, Paris, https://doi.org/10.1787/3d96efc5-en

Giuliano, P. (2017). *Gender: An historical perspective*. NBER Working Paper 23635.

Golden, L., Lambert, S., Henly, J., & Wiens-Tuers, B. (2011). Working time in the employment relationship: Perceived control and work-life balance. In K. T. Wilkinson (Ed.), *Edward Elgar handbook on work and employment relations.* Edward Elgar.

Golden, L. (2011). *The effects of working time on productivity and firm performance: A research synthesis paper.* ILO, conditions of work and employment series no. 33.

Goldin, C. (2014). A grand gender convergence: Its last chapter. *American Economic Review, 104*(4), 1091–1110.

Gomes, P. (2021). *Friday is the new saturday*. The History Press Ltd.

Haraldsson, G. D., & Kellam, J. (2021) *Going Public: Iceland's journey to a shorter working week.* Alda and Autonomy.

Harper, A., & Martin, A. (2018). *Achieving a shorter working week in the UK*. The New Economics Foundation.

Heejung, C. (2022). A social policy case for a four-day week. *Journal of Social Policy, 51*(3), 551–566.

Hunnicutt, B. K. (1992). Kellogg's six-hour day: A capitalist vision of liberation through managed work. *The Business History Review, 66*(3), 475–522.

ILO. (2023). *Statistics on working time*. From ILO Stat: https://ilostat.ilo.org/topics/working-time/

Jahal, T., Bardoel, E. A.., & Hopkins, J. (2023), Could the 4-day week work? A scoping review. *Asia Pacific Journal of Human Resources.* https://doi.org/10.1111/1744-7941.12395

Johnson, J. V., & Lipscomb, J. (2006). Long working hours, occupational health and the changing nature of work organization. *American Journal of Industrial Medicine, 49*(11), 921–9.

Kallis, G., Kalush, M., O'Flynn, H., Rossiter, J., & Ashford, N. (2013). Friday off: Reducing working hours in Europe. *Sustainability (Switzerland), 5*(4), 1545–1567. https://doi.org/10.3390/su5041545

Kamerade, D., Wang, S., Burchelll, B., Balderson, S. U., & Coutts, A. (2019). A shorter working week for everyone: How much paid work is needed for T mental health and well-being? *Social Science and Medicine, 241*, 112353.

Keus Van De Poll, M., Nybergh, L., Lornudd, C., Hagberg, J., Bodin, L., Kwak, L., & Bergstrom, G. (2020). Preventing sickness absence among employees with common mental disorders or stress-related symptoms at work: A cluster randomised controlled trial of a problem-solving-based

intervention conducted by the occupational health services. *Occupational and Environmental Medicine, 77*(7), 454–461.

Kim, S., McLean, G. N., & Park, S. (2018). The cultural context of long working hours: Workplace experiences in Korea. *New Horizons in Adult Education and Human Resource Development, 30*(2), 36–51.

Knight, K., Rosa, E. A., & Schor, J. B. (2012) *Reducing growth to achieve environmental sustainability: The role of work hours* (Issue 304).

Kodz, K., Lain, D., Strebler, M., Rick, J., Bates, P., & Meager, N. (2023). *Working long hours: A review of the evidence* (vol. 1). Employment Relations Research Series ERRS16 | Department of Trade and Industry.

Künn-Nelen, A., De Grip, A., & Fouarge, D. (2013). Is part-time employment beneficial for firm productivity? *Industrial and Labour Relations Review, 66*(5), 1172–1191.

Landrigan, C., Rothschild, J., Cronin, J., Kaushal, R., Burdick, E., Katz, J., & Czeisler, C. (2004). Effect of reducing interns' work hours on serious medical errors in intensive care units. *The New England Journal of Medicine, 351*(18), 1838–48.

Lepinteur, A. (2019). The shorter workweek and worker wellbeing: Evidence from Portugal and France. *Labour Economics, 58*, 204–220.

Lewis, K., & Stronge, W. (2021). *A shorter working week in Government: A worker-led inquiry into desirability, feasibility and impact.* Autonomy.

Lewis, K., Stronge, W., Kellam, J., & Kikuchi, L. (2023) *The results are in: The UK's four-day week pilot*. Autonomy, 4Day Week, 4 Day Week Global.

Li, W., Ruan, W., Chen, Z., Yi, G., & Wang, D. (2021). A meta-analysis of observational studies including dose–response relationship between long working hours and risk of obesity. *Reviews in Endocrine and Metabolic Disorders, 22*, 837–845.

Marston, G. (2016). Greening the Australian welfare state: Can basic income play a role? In *Basic income in Australia and New Zealand: Perspectives from the neoliberal frontier* (pp. 157–177). New York, Palgrave Macmillan US.

Milner, A., Smith, P., & LaMontagne, A. D. (2015). Working hours and mental health in Australia: Evidence from an Australian population-based cohort, 2001–2012. *Occupational and Environmental Medicine, 72*, 573–579.

Moores, J. E. (1990). A meta-analysis of the effects of compressed work weeks. *Applied HRM Research, 1*(1), 8–14.

Mullens, F., & Ignace, G. (2023). Reducing weekly working hours: Temporal strategies and changes in the organization and experiences of work-results from a qualitative study of 30-hour workweek experiment. *Time and Society, 32*(2), 146–168.

Mullens, F., Verbeylen, J., & Glorieux, I. (2021). Rethinking the workweek: Results from a longitudinal time-use study of a 30-hour workweek experiment. *Journal of Time Use Research, 16*(1), 52–75.

OECD. (2021). *The role of firms in wage inequality: Policy lessons from a large scale cross-country study*. OECD Publishing, Paris. https://doi.org/10.1787/7d9b2208-en

OECD. (2022). *Incidence of FTPT employment*. From OECD Stat: https://stats.oecd.org/Index.aspx?DataSetCode=FTPTC_I

OECD. (2023). *Framework for the evaluation of SME and entrepreneurship policies and programmes 2023*. OECD Studies on SMEs and Entrepreneurship, OECD Publishing.

Pang, A. S. (2020). *Shorter: How smart companies work less, embrace flexibility and boost productivity*. Penguin.

Pencavel, J. (2015). The productivity of working hours. *The Economic Journal, 125*(589), 2052–2076.

Piasna, A., & Spiegelaere, S. D. (2021). Working time reduction, work/life balance and gender equality. *Dynamiques Regionales, 10*, 19–42.

Pullinger, M. (2014). Working time reduction policy in a sustainable economy: Criteria and options for its design. *Ecological Economics*, *103*, 11–19. https://doi.org/10.1016/j.ecolecon.2014.04.009

Raposo, P. S., & Ours, J. C. (2010). How working time reduction affects jobs and wages. *Economic Letters, 106*, 61–63.

Rivera, A., Akanbi, M., O'Dwyer, L., & McHugh, M. (2020). Shift work and long work hours and their association with chronic health conditions: A systematic review of systematic reviews with meta-analyses. *PLoS ONE, 15*(4), e0231037.

Rosengren, C. (2019). Performing work: The drama of everyday working life. *Time and Society, 28*(2), 613–633.

Rutherford, S. (2001). Are you going home already? The long hours culture, women managers and patriarchal closure. *Time and Society, 10*(2/3), 259–276.

Sanchis, J. (2023), *Cuatro Días: Trabajar Menos Para Vivir En Un Mundo Mejor.* Barlin Libros.

Schiller, H., Lekander, M., Rajaleid, K., Hellgren, C., Åkerstedt, T., Barck-Holst, P., & Kecklund, G. (2017). The impact of reduced worktime on sleep and perceived stress—a group randomized intervention study using diary data. *Scandinavian Journal of Work, Environment & Health*, 109–116.

Schor, J. B., Fan, W., Kelly, O., Gu, G., Bezdenezhnykh, T., & Bridson-Hubbard, N. (2022). *The four day week: Assessing global trials of reduced work time with no reduction in pay.* Four Day Week Global.

Schwarz, U. V., & Hasson, H. (2011). Employee self-rated productivity and objective organizational production levels: Effects of worksite health interventions involving reduced work hours and physical exercise. *Journal of Occupational and Environmental Medicine, 3*(8), 838–844.

Schwarz, U. V., Lindfors, P., & Lundberg, U. (2008) Health-related effects of worksite interventions involving physical exercise and reduced workhours. *Scandinavian Journal of Work, Environment & Health, 34*(3), 179–188.

Shangguan, R., Devaro, J., & Owan, H. (2021) *Enhancing team productivity through shorter working hours: Evidence from the great recession.* RIETI Discussion Paper Series 21-E-040.

Shepard, E., & Clifton, T. (2000). Are longer hours reducing productivity in manufacturing? *International Journal of Manpower, 21*(7), 540–553.

Sparks, K., Faragher, B., & Cooper, C. (2001). Well-being and occupational health in the 21st century workplace. *Journal of Occupational and Organizational Psychology, 74*(4), 489–509.

Specchia, G. L., & Vandenberghe, V. (2013). Is part-time employment a boon or bane for firm productivity? *Unpublished paper*.

Spencer, D. A. (2022). A four-day working week: Its role in a politics of work. *The Political Quarterly, 93*(3), 401–407.

Spiegelaere, S. D., & Piasna, A. (2017). *The why and how of working time reduction.* European Trade Union Institute.

Stagl, S. (2014). Ecological macroeconomics: Reflections on labour markets. *European Journal of Economics and Economic Policies: Intervention, 11*(2), 171–181. https://doi.org/10.4337/ejeep.2014.02.04

Stronge, W., & Harper, A. (2019). *The shorter working week: A radical and pragmatic proposal*. Autonomy.

US Bureau of Labor Statistics. (2023). *Unemployment rate, FRED Sr. Louis Fed*. From https://fred.stlouisfed.org/series/UNRATE

Veal, A. J. (2023). The 4-day work-week: The new leisure society? *Leisure Studies, 42*(2), 172–187.

Virtanen, M., Jokela, M., & Lallukka, T. (2022). Long working hours and change in body weight: Analysis of individual-participant data from 19 cohort studies. *International Journal of Obesity, 44*, 1368–1375.

Voglino, G., Savatteri, A., Gualano, M., Catozzi, D., Rousset, S., Boietti, E., & Siliquini, R. (2022). How the reduction of working hours could influence health outcomes: A systematic review of published studies. *BMJ Open, 12*.

Weeden, K. A., Cha, Y., & Bucca, M. (2016). Long work hours, part-time work, and trends in the gender gap in pay, the motherhood wage penalty, and the fatherhood wage premium. *RSF: The Russell Sage Foundation Journal of the Social Sciences*, *2*(4), 71–102.

Williams, J. (2010). *Reshaping the work-family debate: Why men and class matter*. Harvard University Press.

Wong, K., Chan, A. H., & Ngan, S. (2019). The effect of long working hours and overtime on occupational health: a meta-analysis of evidence from 1998 to 2018. *International Journal Environmental Research and Public Health, 16*(12), 2102.

Worstall, T. (2019). *The problem with the four day work week*. 1–2. https://www.adamsmith.org/blog/the-problem-with-the-four-day-work-week

Zheng, H., Vatsa, P., Ma, W., & Zhou, X. (2023). Working hours and job satisfaction in China: A threshold analysis. *China Economic Review, 77*, 101902.